

Package ‘cic’

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Title Changes-in-Changes Estimator with Asymptotic Confidence Intervals

Version 0.1.0

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Description Implements the Changes-in-Changes estimator and asymptotic confidence intervals for empirical quantile-based estimators. defined as the average of the quantile-transformed ranks. Provides asymptotic confidence intervals via a full-sample (no-split) variance estimator and bootstrap alternatives (percentile and standard-error). The core computations are written in C++ via Rcpp for performance. Includes a comprehensive diagnostic tool, `check_cic_assumptions()`, that validates empirical data against theoretical assumptions (sampling balance, continuity, tail behavior, and convergence rates). Features robust input validation, graceful error handling for missing compiled code, and pure R fallback implementations for maximum compatibility.

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Imports Rcpp (>= 1.0.0)

Suggests testthat (>= 3.0.0), knitr, rmarkdown

LinkingTo Rcpp

VignetteBuilder knitr

Config/testthat/edition 3

NeedsCompilation yes

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check_cic_assumptions *Check CiC Model Assumptions*

Description

Validates that data satisfy the theoretical requirements for the CiC estimator.

Usage

```
check_cic_assumptions(Y, X, Z)
```

Arguments

Y	Numeric vector: Outcome variable
X	Numeric vector: Treatment/endogenous variable
Z	Numeric vector: Instrument/exogenous variable

Value

A list with elements:

pass_all	Logical: TRUE if all checks passed
metrics	Named list of diagnostic metrics: • lambda1, lambda2, lambda3 Sample size ratios • d1_left_tail, d2_right_tail Tail indices • b1_left_boundary, b2_right_boundary Boundary density estimates • autocorrelation_pLjung-Box test p-value • uniqueness_ratio Ratio of unique values
messages	Character vector: Warnings or success messages

cic *Changes-in-Changes Estimator*

Description

Computes the CiC estimator for nonlinear difference-in-differences models as described in the manuscript by Chhor, D'Haultfoeuille, L'Hour, and Mugnier.

Usage

```
cic(
  Y,
  X,
  Z,
  method = c("no-split", "split", "kde", "bse", "bpc"),
  B = 200,
  h = NULL,
  level = 0.95
)
```

Arguments

<code>Y</code>	Numeric vector: Outcome variable
<code>X</code>	Numeric vector: Treatment/endogenous variable
<code>Z</code>	Numeric vector: Instrument/exogenous variable
<code>method</code>	Character vector: Estimation method(s). Options are: • "no-split" Nonparametric method using full sample (fastest) • "split" Sample-splitting variance estimator • "kde" Epanechnikov KDE variance estimator • "bse" Bootstrap standard-error method • "bpc" Bootstrap percentile method
<code>B</code>	Integer: Number of bootstrap replications (default: 200). Only used for "bse" and "bpc".
<code>h</code>	Numeric: Bandwidth parameter for no-split method (default: NULL = auto)
<code>level</code>	Numeric: Confidence level for intervals (default: 0.95)

Value

An S3 object of class 'cic' with elements:

<code>theta_hat</code>	Estimated CiC parameter
<code>ci</code>	Data frame with confidence intervals (columns: lower, upper, length, method)
<code>n</code>	Sample sizes (Y, X, Z)
<code>method</code>	Estimation method(s) used
<code>h</code>	Bandwidth used for the nonparametric density estimators
<code>level</code>	Confidence level

References

Chhor, J., D'Haultfoeuille, X., L'Hour, J., & Mugnier, M. (2026). Asymptotic Properties of Empirical Quantile-Based Estimators. Manuscript.

See Also

[check_cic_assumptions](#), [sim_dgp](#)

Examples

```
set.seed(42)
d <- sim_dgp(500)
fit <- cic(d$Y, d$X, d$Z, method = "no-split")
summary(fit)
```

qY_dgp	<i>Quantile Function for the CiC Simulation DGP</i>
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Description

Quantile function from the Monte Carlo section of the CiC simulation design:

$$F_Y^{-1}(t) = -t^{-d1} + (1-t)^{-d2}$$

with the convention that $d1 = 0 \Rightarrow t^{-d1} = 1$, $d2 = 0 \Rightarrow (1-t)^{-d2} = 1$.

Usage

```
qY_dgp(t, d1 = 0, d2 = 0.05)
```

Arguments

t	Numeric vector in $[0, 1]$.
d1	Left tail parameter (default: 0)
d2	Right tail parameter (default: 0.05)

Value

Numeric vector of quantiles

Examples

```
qY_dgp(0.5, d1 = 0, d2 = 0.05)
```

sim_dgp	<i>Simulate Data from the CiC Simulation DGP</i>
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Description

Simulates a dataset from the Data Generating Process (DGP) used in the package examples and Monte Carlo checks.

Usage

```
sim_dgp(n, b1 = 0, b2 = 0.05, d1 = 0, d2 = 0.05, seed = NULL)
```

Arguments

n	Sample size
b1	Left boundary parameter (default: 0)
b2	Right boundary parameter (default: 0.05)
d1	Left tail parameter (default: 0)
d2	Right tail parameter (default: 0.05)
seed	Random seed (default: NULL)

Value

A list with elements:

Y	Outcome variable
X	Treatment/endogenous variable
Z	Instrument/exogenous variable

Examples

```
set.seed(42)
d <- sim_dgp(500)
head(d)
```

theta_true	<i>True CiC Parameter for the Simulation DGP</i>
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Description

Computes the true theta_0 parameter for the DGP:

$$\theta_0 = [B(1 - b_1, 1 - b_2 - d_2) - B(1 - b_1 - d_1, 1 - b_2)] / B(1 - b_1, 1 - b_2)$$

where B is the beta function.

Usage

```
theta_true(b1 = 0, b2 = 0.05, d1 = 0, d2 = 0.05)
```

Arguments

b1	Left boundary parameter (default: 0)
b2	Right boundary parameter (default: 0.05)
d1	Left tail parameter (default: 0)
d2	Right tail parameter (default: 0.05)

Value

Numeric scalar: true theta_0 value

Examples

```
theta_true(b1 = 0, b2 = 0.05, d1 = 0, d2 = 0.05)
```

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